Could we have seen it coming?
A pre-landing hazard re-analysis of the IM-2 "Athena" lunar south-pole touchdown

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HATI v2.0 – Máni Mission terrain-hazard pipeline

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Abstract

In March 2025 the Intuitive Machines IM-2 lander ("Athena") reached the lunar south pole and toppled near a small crater on sloped ground. The post-landing attribution is clear; the pre-landing question is not. Could the danger have been read off maps that already existed? I test this with HATI, a two-channel terrain pipeline, using only data acquired in 2012, thirteen years before the landing. Channel one builds a multi-scale roughness heatmap from a 4 m/px stereo elevation model. Channel two counts the cast shadows of obstacles in a 0.9 m/px photograph, reaching sizes the elevation model cannot resolve. Read at the published touchdown coordinate, the heatmap ranks the point in the worst 3.7 % of an already-rough massif (robust $z = 3.22$ above the scene median), and it does so while the local slope is a gentle 2.8°: the flag is texture, not tilt. A per-channel decomposition shows the gross-tilt channels voting the site *safe* while the feature-scale channels overrule them, which is exactly the failure mode a slope-only screen misses. The shadow census finds ∼1400 obstacles per km^2 below the elevation model's 8 m floor. Both results come from first-principles geometry, with no photometric scattering model in the pipeline. I argue the site was detectable from pre-landing data, state plainly what this does and does not prove, run a within-scene safe-ground control that names the discriminating channels, and lay out the two control experiments that would turn "worst few percent here" into an absolute claim.

Keywords: lunar landing hazards; sub-resolution roughness; LROC NAC; shadow photoclinometry; Mons Mouton; terrain-relative navigation.

1 Introduction

On 6 March 2025 the IM-2 spacecraft set down at roughly 84.79° S, 29.20° E on Mons Mouton, deep in the lunar south-polar highlands, and tipped over. The Lunar Reconnaissance Orbiter Camera (LROC) team later tied the upset to a crater about 20 m across sitting on sloped ground at the touchdown.

That is the hindsight story. This paper asks a different, harder question, and tries to answer it without cheating: *if we had run a dedicated terrain-hazard analysis on the maps that already existed before the landing, would it have pointed at that exact spot?*

Two rules keep the test honest. First, every input predates the landing by thirteen years, so nothing in the pixels could have been shaped by the crash. Second, the landing *outcome* never enters the math; the only thing borrowed from after the fact is the single coordinate telling me which pixel to read, and that pixel is scored the same way as every other one.

Why it is built this way

A "what if we had looked earlier" study is dangerous precisely because the answer is already known. It is easy, even unconsciously, to tune an analysis until it agrees with reality. The two guardrails above (pre-landing-only inputs, outcome-free scoring) are there so the result

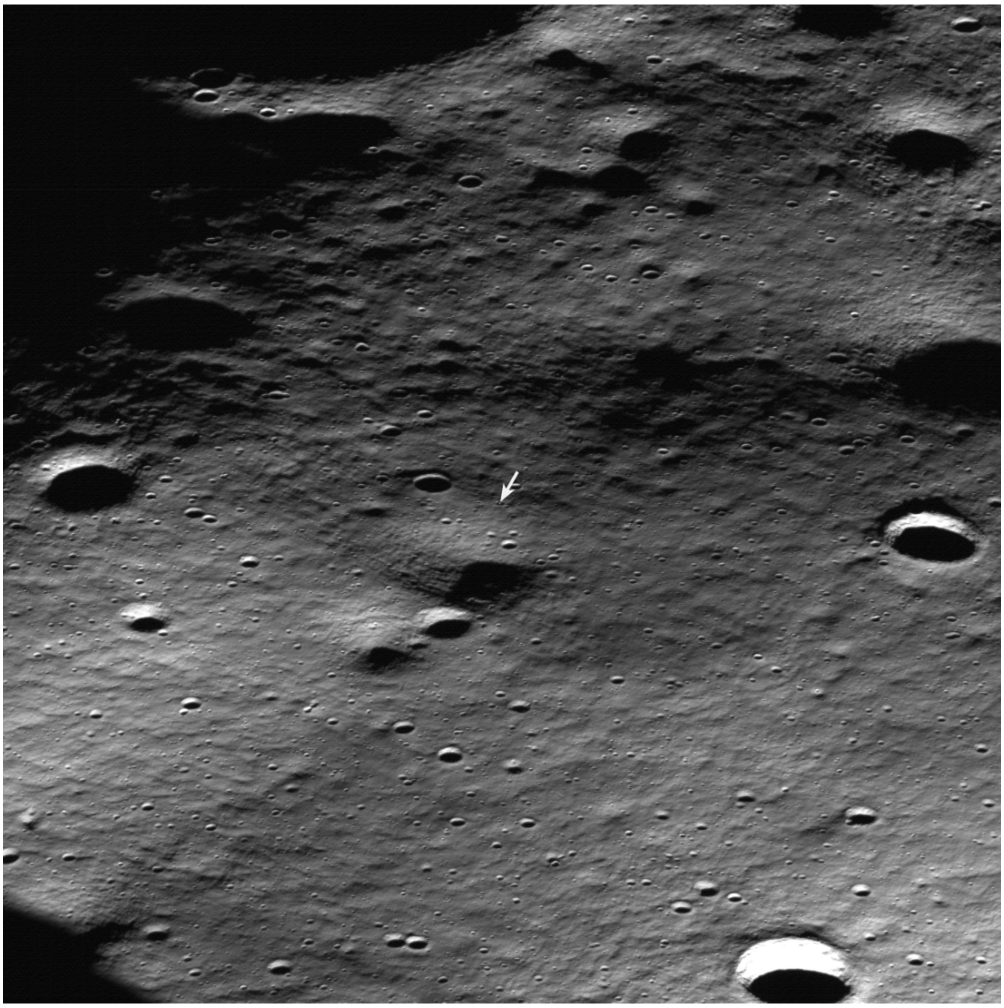

Figure 1: The IM-2 Athena landing site on Mons Mouton, lunar south pole, in an LROC NAC oblique view acquired after the landing. The grazing polar sunlight that throws the long shadows here is the same illumination that makes Channel 2 work. Base image: NASA/GSFC/Arizona State University, LROC team.

is a measurement, not a self-fulfilling prophecy. This is a *detectability* test. It is not a claim that HATI would have changed the mission, and it is not a blind prediction, because I knew where to look.

HATI runs two independent channels, sketched in fig. 2. The contribution of this paper is to run both on the real failure site, decompose exactly why the flag fires, and fence the result with the limitations a reviewer would raise.

2 Data, and the rule against cheating

Both maps come from the LROC NAC stereo product `NAC_DTM_NOBILE03` (Arizona State University; Robinson et al. 2010), built from three NAC frames acquired on 31 August 2012:

- an elevation model (DTM) at 4.0 m/px, 1473×1172 pixels;
- a co-registered orthophoto at 0.9 m/px, 6547×5209 pixels, 16-bit.

Why it is built this way

The choice of a 2012 product is the whole guardrail. Anything built from post-landing imagery would let the crash leak into the "prediction." Using source frames from thirteen years earlier means the test is fair: the maps knew nothing about IM-2.

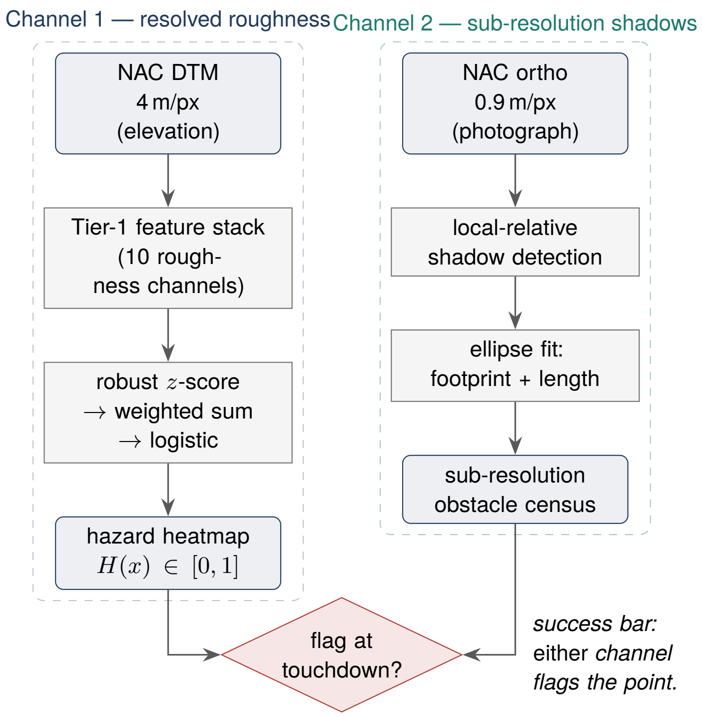

Figure 2: The two HATI channels. Channel 1 turns the elevation model into a ranked hazard score; Channel 2 turns the photograph into a census of obstacles too small for the elevation model to see. The test passes if either one flags the touchdown.

3 Finding the exact spot

Both maps use a south-polar stereographic grid on a sphere of radius $R = 1737.4\,\text{km}$, centred on the pole, longitude measured east (fig. 4). To convert the touchdown latitude φ and longitude λ into a position on that grid I use the closed-form spherical projection,

$$\rho = 2R\,\tan\left(\frac{\pi}{4} + \frac{\varphi}{2}\right), \qquad x = \rho \sin\lambda, \qquad y = \rho\cos\lambda, \tag{1}$$

then divide by the pixel size and offset by the map corner. I ran this two independent ways, eq. (1) and the GDAL coordinate engine, and they land on the same pixel with zero disagreement: row 1181, column 387 on the elevation model, row 5248, column 1721 on the photograph.

In plain words

A stereographic map is what you get if you stand at the north pole and trace every south-polar point onto a flat sheet. Near the pole it barely distorts anything, so distances and shapes stay honest. Equation (1) is just the rule for "how far from the centre, and in which direction" for a given latitude and longitude.

4 Channel 1: the roughness heatmap

4.1 The idea

In plain words

Picture a parking lot shot from so high up that anything smaller than a car blurs into a smudge. You cannot see individual potholes. But a patch full of potholes still *looks* different from smooth asphalt: speckled, busy, high-variance. The heatmap is a machine for measuring that busy-ness at several sizes at once, then boiling it down to one hazard number between 0 and 1. A 20 m crater is only five pixels wide on a 4 m map, so you will not always see the

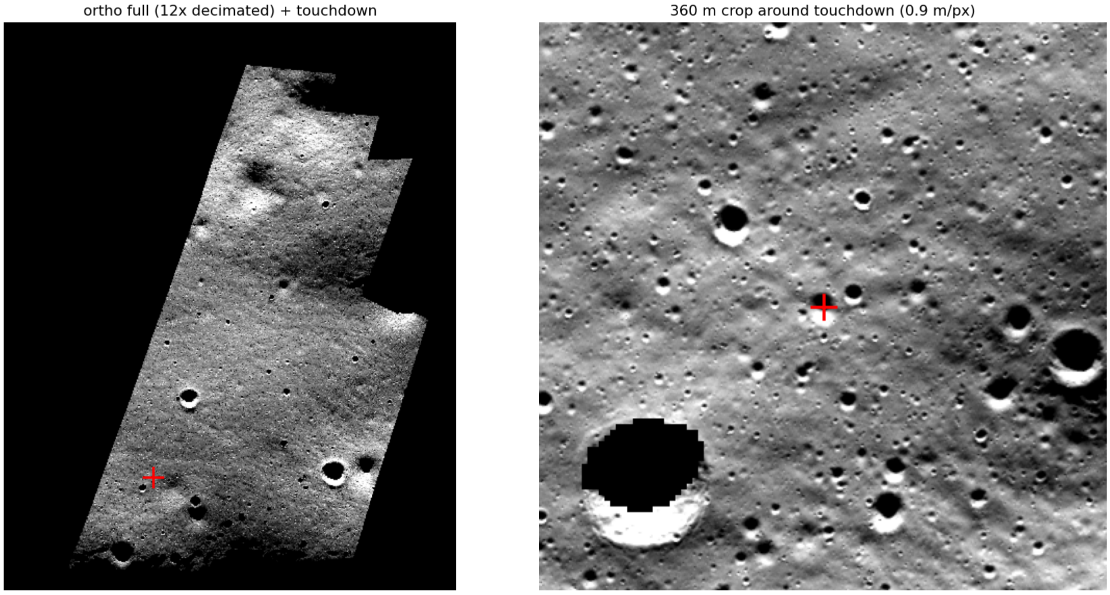

Figure 3: Left: the full 0.9 m/px orthophoto (the data covers a tilted diamond inside the rectangular file; black is outside coverage), with the touchdown marked. Right: a 360 m crop around the touchdown. The point sits in a dense field of small craters, several of which are far below the elevation model's resolving limit. Note the pitch-black, fully shadowed interior of the large crater at lower left: grazing polar light turns relief into long, readable shadows.

bowl, but you will see the neighbourhood get statistically rougher. That is the signal.

4.2 The channels, why each one is included, and how hard each one pushed

Every channel is an image the size of the elevation model, where bright means "rough here." They are built from two primitives. Slope is rise over run,

$$s(x,y) = \arctan\sqrt{\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2}, \tag{2}$$

and curvature is the Laplacian $\nabla^2 z = \partial^2 z/\partial x^2 + \partial^2 z/\partial y^2$, positive in a bowl, negative on a dome. Each channel measures one of these inside a small sliding window.

The cleverest family is Median Differential Slope (MDS). It blurs the terrain to two sizes, L and $2L$, measures slope on each, and subtracts. What survives is the roughness that lives specifically around size L. Run it at several L and you get a little size-spectrum of bumpiness; I use $L \approx 8, 12, 20, 32$ m, which brackets the crater Athena struck (Kreslavsky and Head 2000; Rosenburg et al. 2011).

Table 1 is the heart of the proof of concept. It decomposes the score *at the touchdown*: each channel's z-score there, its literature weight, and its additive contribution to the final logit. The contributions sum, the bias -1.0 is added, the logistic squashes the total, and the result is $H = 0.726$. The same data is drawn in fig. 5.

Each channel, with the reason it earns a place:

Slope IQR (spread of slope directions; weight 1.00; biggest driver, $+0.476$). A clean plane has slopes all pointing one way, so the spread is tiny; a crater field has slopes pointing everywhere, so the spread explodes (Kreslavsky et al. 2013). At the touchdown it maxed out at the $+4$ clip. Fingerprint of a pitted surface.

MDS @ 12 m (roughness at the 12 m size; weight 0.90; $+0.406$). Differential slope isolates bumpiness at one chosen size, and 12 m sits right on the crater the LROC team blamed. This channel put its finger on the hazard's actual size.

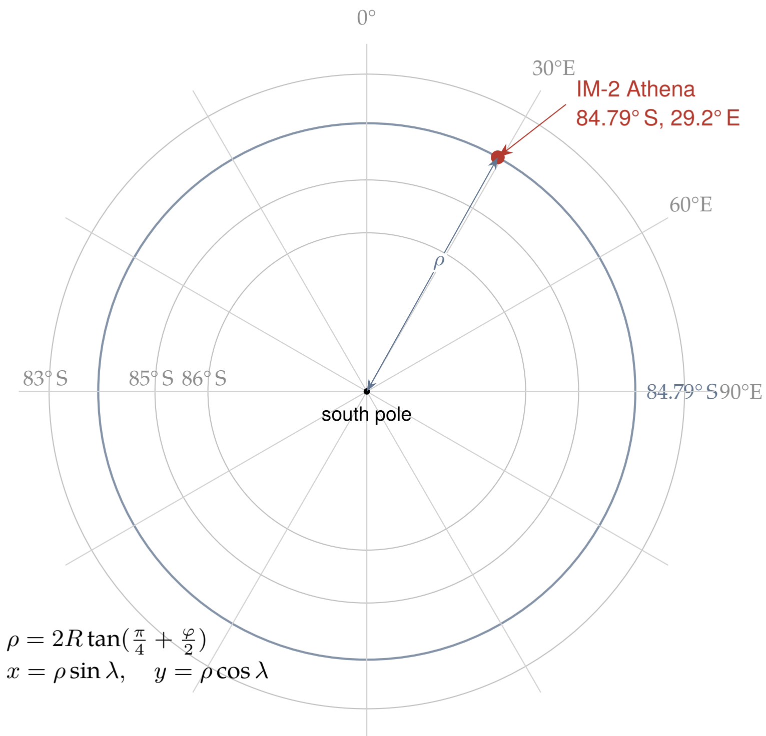

Figure 4: Turning a latitude/longitude into a pixel. Looking straight down on the south pole, lines of latitude become circles and lines of longitude become spokes. A point's distance from the pole, ρ, depends only on its latitude; the spoke it sits on is its longitude. The touchdown (red) is 5.21° of latitude from the pole, on the 29.2° E spoke.

|**TPI**| (height above/below the local mean; weight 0.70; $+0.333$). A lander cares whether it is perched on a rim or sunk in a bowl. At the 99th percentile, the point sits on strong local relief.

MDS @ 8 m and @ 20 m (weights 1.00, 0.80; $+0.321$, $+0.278$). The smaller MDS sizes sit nearest the resolution floor, where an unresolved hazard leaves its first mark, so they carry top weight (Cai and Fa 2020). Both fired hard: the danger has structure across the whole 8–20 m band.

Curvature IQR and Planar deviation (weights 1.00, 0.70; $+0.153$, $+0.134$). Craters carry a telltale curvature pattern (positive rim, negative floor); planar deviation catches general lumpiness. Both moderately raised, backing the cratered read without running the show.

MDS @ 32 m (weight 0.60; almost nothing, $+0.030$). The coarsest size here. Its near-silence is itself a finding: the hazard is smaller than 32 m.

RMS slope and TRI (average slope, single-step jaggedness; weights 1.00, 0.70; both **negative**). And here is the line that makes the proof of concept land (Rosenburg et al. 2011; Riley et al. 1999). Both sit *below* the scene median. The average tilt is gentle and the pixel-to-pixel steps are small. On their own, these two channels call the site calm.

The hazard signature, in one sentence

Low average slope, low jaggedness, but maxed slope spread, near-maxed TPI, and high MDS across 8–20 m. In plain words: gently tilted and smooth between features, yet packed with crater-scale relief. That is what a gently-sloped, heavily-cratered patch looks like, and it is exactly the case a slope-only screen waves through (fig. 6). The flag is not one channel shouting; it is the disagreement between the channels that read gross tilt and the channels that read structured roughness, settled inside a single number, from topography alone.

Table 1: Per-channel decomposition at the touchdown (config: window 7, MDS 8–32 m). "Contribution" is each channel's additive share of the pre-logistic score. A positive value pushes the hazard up.

Channel	Weight	z at touchdown	Contribution	Value pct.	Push
Slope IQR	1.00	4.00	0.476	98.7	up
MDS @ 12 m	0.90	3.79	0.406	98.4	up
|TPI|	0.70	4.00	0.333	99.0	up
MDS @ 8 m	1.00	2.69	0.321	96.0	up
MDS @ 20 m	0.80	2.92	0.278	96.7	up
Curvature IQR	1.00	1.29	0.153	85.0	up
Planar deviation	0.70	1.60	0.134	88.9	up
MDS @ 32 m	0.60	0.42	0.030	64.8	up (weak)
RMS slope	1.00	−0.64	−0.077	24.7	**down**
TRI	0.70	−0.94	−0.079	16.2	**down**

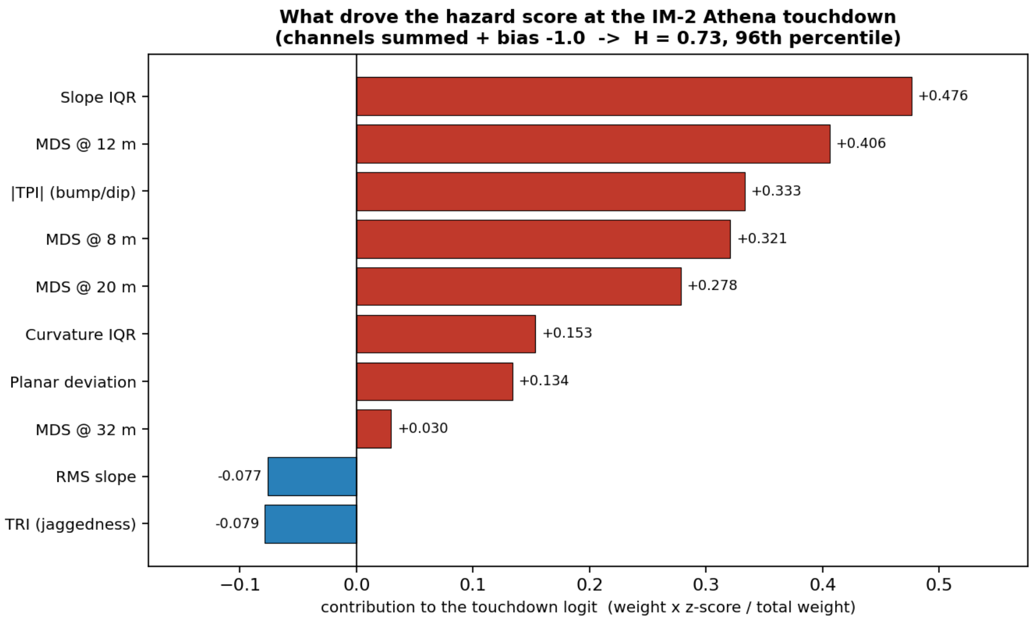

Figure 5: What drove the hazard score at the touchdown. Red bars push the score up, blue bars pull it down. The two channels that read *average* tilt (RMS slope, TRI) vote the site safe; the channels that read *structured, feature-scale* roughness overrule them.

4.3 Filling the holes in the map

The elevation file is a rectangle, but the real data fills only a tilted diamond inside it; about 45 % is "no data." Sliding-window math across the cliff between real data and a hole produces garbage spikes.

Why it is built this way

I fill every hole with the value of its *nearest real pixel* before any window math, so the filled region is a smooth continuation instead of a cliff. Then, after computing the channels, I discard every pixel close enough to a hole that a window could have peeked in. The touchdown sits 116 m from the nearest hole, far enough that the 8–32 m channels never reach the edge.

4.4 Folding ten channels into one number

Each channel is first normalised by a robust z-score, using the median and the inter-quartile range (IQR) rather than the mean and standard deviation, after a $\log(1 + \cdot)$ squash and a clip at

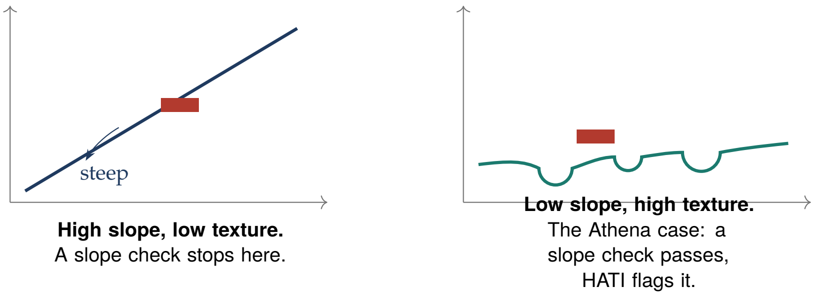

Figure 6: Why "low slope" is not "safe." A steep clean ramp (left) is caught by an ordinary slope check. A gently-tilted but crater-pocked surface (right) passes that check while being genuinely dangerous. Channel 1 is built to separate these two.

Table 2: Robustness sweep. Every configuration that can be *computed* at this near-edge site puts the touchdown in the worst 4–7 %. The Apollo-default config returns "undefined" because its 80 m scale reaches the data edge 116 m away.

Configuration	MDS sizes	H at touchdown	Scene percentile	Flag (>p75 / p90)
Apollo default (win. 11)	12–80 m	undefined	—	—
window 5	8–32 m	0.757	97.1	yes / yes
window 7	8–32 m	0.726	96.3	yes / yes
window 9	8–24 m	0.645	93.3	yes / yes
window 7	8–20 m	0.780	97.4	yes / yes

± 4:

$$z_k = \mathrm{clip}_{\pm 4}\left(\frac{g(c_k) - \mathrm{med}\ g(c_k)}{\mathrm{IQR}\, g(c_k)/1.349}\right), \qquad g(\cdot) = \log(1+\cdot). \tag{3}$$

Then a literature-weighted sum, normalised by the total weight, shifted by a bias, and squashed to $[0, 1]$ by the logistic σ:

$$H = \sigma\left(\frac{\sum_k w_k z_k}{\sum_k |w_k|} + b\right), \qquad \sigma(t) = \frac{1}{1+e^{-t}}, \qquad b = -1.0. \tag{4}$$

Why it is built this way

Three deliberate choices. The median/IQR (not mean/standard-deviation) means a few freak pixels cannot hijack the scale. The $\log(1+\cdot)$ tames the long tail of roughness so flat terrain still spreads out usefully. Dividing by the total weight decouples the score from the number of channels, so adding or removing one does not silently move the midpoint. The weights come from the roughness literature, not from tuning against Athena.

Honest limit

H is a *relative* score: it ranks each pixel against the rest of this scene. "96th percentile" means rougher than 96 % of the Mons Mouton massif, and that massif is rough everywhere. Turning a ranking into an absolute hazard needs a calibration baseline; see section 9.

4.5 Reading the score, without fishing for it

The honest worry with any multi-knob method is that the knobs were turned until the site looked guilty. To kill that worry I swept window sizes and MDS bands and report *all* of them, including the one that fails (table 2).

At the touchdown the score is $H = 0.726$, the 96.3rd percentile; only 3.7 % of the valid terrain scores rougher, which is 3.22 robust standard deviations above the median. Meanwhile the slope

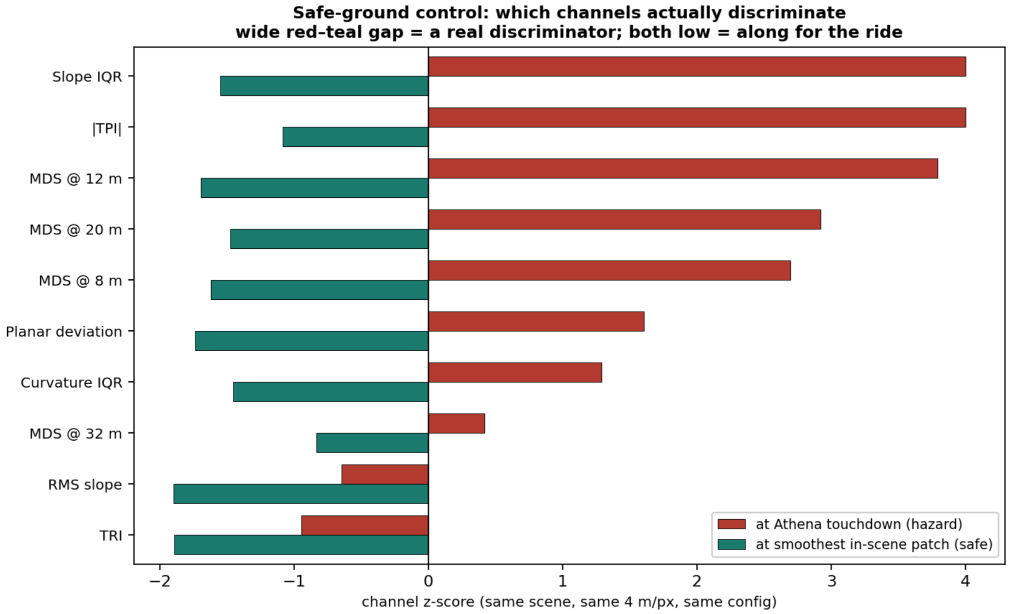

Figure 7: The safe-ground control. For each channel, the z-score at the touchdown (red) and at the smoothest in-scene patch (teal). A wide gap means the channel separates hazard from safe ground; two low bars mean it does not. Slope IQR, the MDS 8–20 m family and |TPI| carry the discrimination; RMS slope and TRI barely move.

there is 2.8° against a scene median of 5.1°: the point is in the *15.9th* percentile for slope, smoother than 84 % of the area. Smoother than most of the site by tilt, rougher than almost all of it by texture.

4.6 Safe-ground control: which channels truly discriminate

A flag means something only if the same channels stay quiet over safe ground. To test that without leaving the scene, I contrast the touchdown against the smoothest patch in the *same* NOBILE03 model, 0.62 km away: there $H = 0.072$ and the slope is 0.9°, against the touchdown's $H = 0.726$ and 2.8°. Product, resolution and scene normalisation are all held fixed, so the only thing that changes between the two points is the terrain, and their z-scores are directly comparable.

Why it is built this way

Why not run the pipeline on a separate mare scene, the obvious "safe control"? Two traps. Roughness is scale-dependent, so a control model at a different pixel size (the 1.5 m/px Apollo 17 model on disk, say) would compare different physical scales. And the z-score is normalised per scene, so scores from two separate runs are not on the same ruler. Holding scene, product and resolution fixed removes both. An external mare control is still worth doing, but only if it matches resolution and compares raw (or commonly-normalised) values, not per-scene z.

This is the "trained eye." Ranked by discrimination $z_{\text{hazard}} - z_{\text{safe}}$, the leaders are Slope IQR ($+5.6$), MDS @ 12 m ($+5.5$), |TPI| ($+5.1$), and MDS @ 20 m and 8 m ($+4.4$, $+4.3$). At the bottom sit RMS slope ($+1.3$) and TRI ($+0.9$), both *below* the scene median at the hazard *and* at safe ground. The channels that do the real work read variability and feature-scale structure; the gross-tilt channels are along for the ride. That is the empirical signature separating safe terrain from unsafe, and it is what a single positive site can legitimately deliver: not just "HATI flagged Athena," but which measurements made it flag.

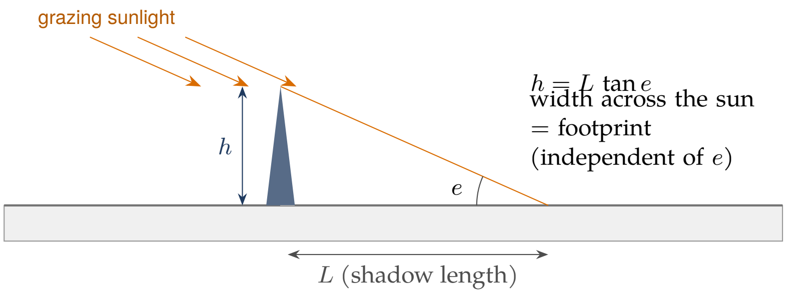

Figure 8: Shadow geometry. The cast shadow's length L converts to the object's height through the solar elevation e (Eq. 5); the width across the sun direction is the object's footprint and does not depend on e. Lower sun means longer shadows, so smaller objects become visible, which is why the south pole is the best place for this method.

Honest limit

The in-scene reference is the smoothest *highland* patch, not a mare, so this shows which channels discriminate *within* this terrain, not the absolute safe-ground floor. It is also picked as the lowest-H patch, so the discrimination ranking is partly self-consistent by construction; an independently chosen flat reference, or the external resolution-matched mare control (section 9), would remove that coupling. The ranking should hold regardless, because RMS slope and TRI sit low at the hazard *too* ($z < 0$), which the choice of a low-H reference does not force.

5 Channel 2: the shadow census

5.1 The idea, and the geometry of a shadow

In plain words

The elevation model stops at 4 m. The photograph is finer than 1 m, and at the south pole the sun barely clears the horizon, so even a knee-high rock throws a long shadow. Those shadows are free measurements of objects far too small to appear on the elevation model.

A shadow gives two separate numbers (fig. 8). Its width *across* the sun is the object's true footprint, independent of how low the sun is. Its length *along* the sun gives the object's height through

$$h = L \tan e, \tag{5}$$

where e is the solar elevation. For this paper the footprint is the headline, because it needs no assumption about the sun angle.

5.2 Why the obvious method fails at the pole

The naive detector calls any dark pixel a shadow. At the pole that fails badly, because half the darkness is not cast shadow at all, it is slopes tilted away from the grazing sun. A global brightness threshold (Otsu) labels 54 % of the crop "shadow," which is meaningless.

Why it is built this way

The fix is to define a shadow *relative to its surroundings*. I build a local brightness baseline with a 40 m median filter, then call a pixel a shadow only if it is darker than half that local baseline. This adapts to the overall light gradient and isolates genuine cast shadows; the dark fraction drops to a sensible 5 % and, as fig. 9 shows, it lights up crater interiors rather than gentle slopes.

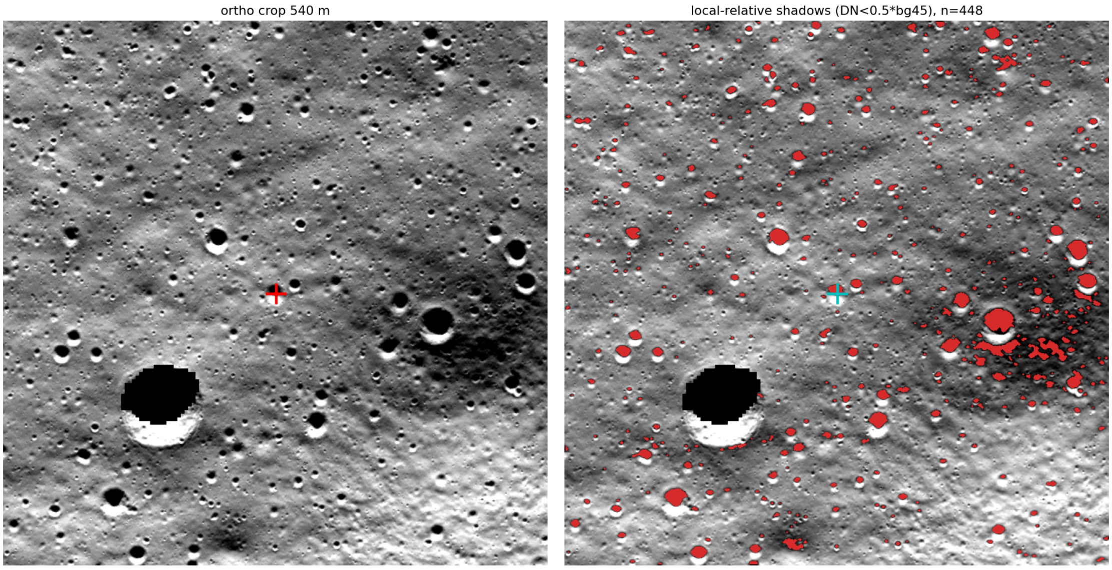

Figure 9: Left: the orthophoto crop. Right: detected cast shadows (red) from the local-relative method. The detector picks out the large crater's shadowed interior and the small craters throughout, while ignoring the broad slope shading. The touchdown (cyan) sits in the obstacle field.

5.3 From dark blobs to measured obstacles, and the resolution floor

After thresholding I clean the mask (one morphological opening to kill speckle, one closing to seal pinholes), label the connected blobs, and fit an ellipse to each one of at least 4 pixels. The short axis is the cross-sun footprint, the long axis is the shadow length. By the sampling theorem (Shannon 1949), the 4 m elevation model cannot register anything below $2 \times 4\,\mathrm{m} = 8\,\mathrm{m}$, so I split every obstacle at that floor.

5.4 A built-in lie detector, and the census

If these "shadows" were random noise their orientations would point everywhere. Real cast shadows all lean away from the sun, so they should share one direction. The measured dominant orientation is 85°, sharply single-peaked: the signature of illumination-driven shadows, not albedo splotches.

Over a 1.08 km^2 box around the touchdown the census finds 1639 shadows ($\sim$1514 per km^2), of which 93.6 % have footprints below the 8 m floor, roughly 1417 per km^2 of obstacles the elevation model cannot resolve, with a median footprint of 3.2 m. Within 25 m of the touchdown there are 3 such sub-resolution obstacles, 9 within 50 m.

Honest limit

The touchdown is *not* a local density spike: its obstacle density is the median for this region. The reason is almost worse than a spike: the whole site is uniformly carpeted with sub-resolution obstacles, so there was no quiet pocket to retarget into. Channel 2 therefore does not independently single out the pixel. It measures a thick hazard layer the elevation model never showed anyone. The shadow count also scales with the threshold ($\sim$twofold), so densities are an order of magnitude, $\sim 10^3\,1/\mathrm{km}^2$, not a precise figure.

6 Both channels together

Figure 10 puts the whole test on one page. Channel 1 ranks the exact touchdown in the worst 4 % of the area, on texture rather than tilt. Channel 2 shows the same area saturated with obstacles below the planning resolution. One pinpoints the spot; both agree the site was dangerous at the

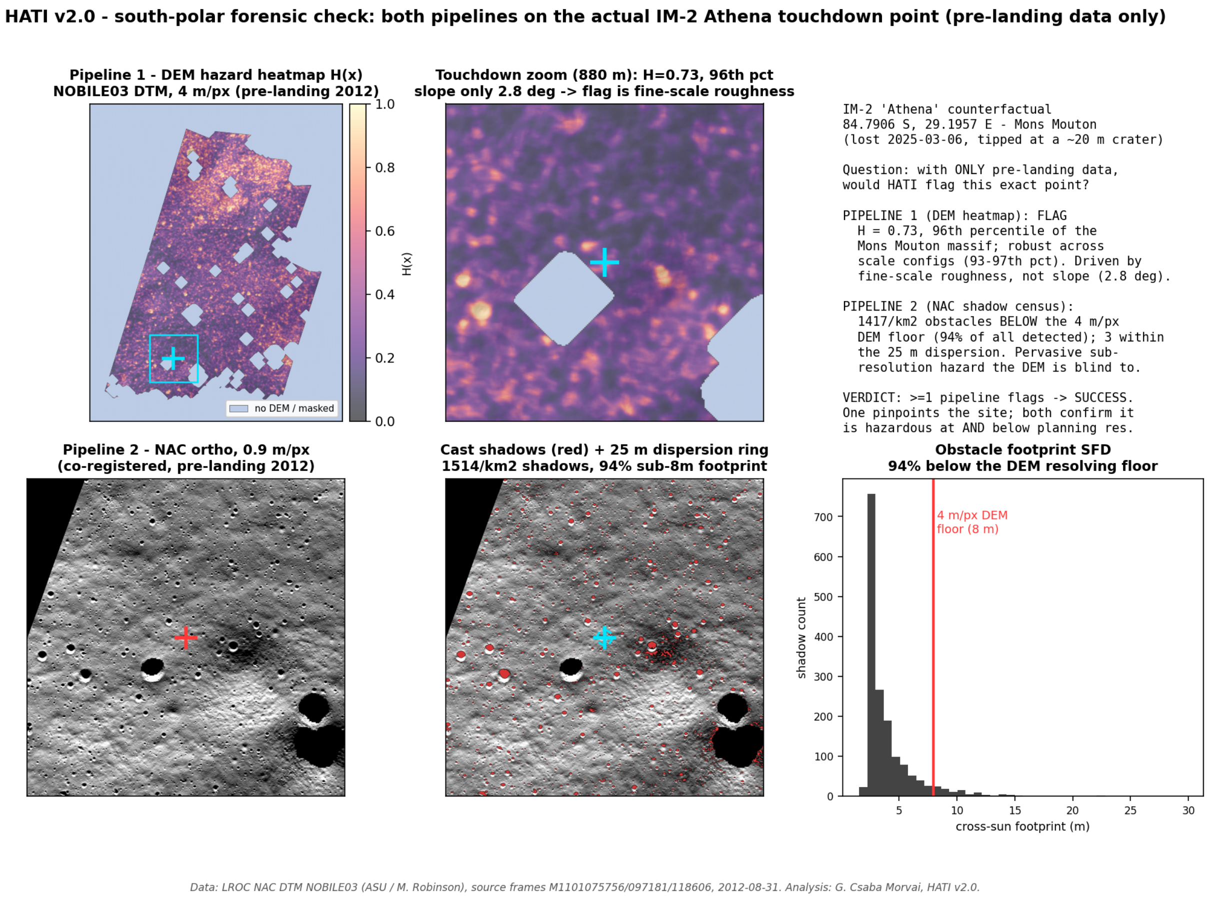

Figure 10: The full counterfactual. Top row, Channel 1: the hazard heatmap over the massif, a zoom on the touchdown ($H = 0.73$, 96th percentile, slope only 2.8°), and the verdict. Bottom row, Channel 2: the orthophoto, the detected shadows with the 25 m dispersion ring, and the obstacle footprint distribution with 94 % below the 8 m floor. Light-slate areas in the top row are no-DEM/masked regions, not terrain.

scale the mission could see and at the scale it could not.

The success bar was: if either channel flags the point, the test passes. Channel 1 flags it. The test passes. I do not upgrade this to "both channels flag it," because Channel 2's density at the point is median, and saying otherwise would be dishonest.

7 The channel we did not build: Hapke photometric roughness ($\bar{\theta}$)

HATI has no photometric-roughness channel. The Hapke scattering model carries a parameter $\bar{\theta}$ ("theta-bar," an *angle*, the mean photometric slope), which measures roughness *below* the size of a single pixel by reading how sunlight scatters across different sun-and-camera angles (Bruce Hapke 1984). The state of the art for the Moon is the near-global $\bar{\theta}$ map of Sato et al. (2014), built from LROC Wide Angle Camera photometry, but binned at $1^\circ \times 1^\circ$ tiles, roughly 30 km on a side. HATI does not compute $\bar{\theta}$ at all, and yet the hazard still shows up through two channels of pure geometry.

In plain words

The thing to be proud of, stated precisely: the hazard signal does not depend on the hard, model-heavy machinery. It falls out of first-principles geometry, through two independent routes that agree. When a result survives that kind of triangulation you trust it more, because two unrelated physical mechanisms would have to fail the same way to fool you.

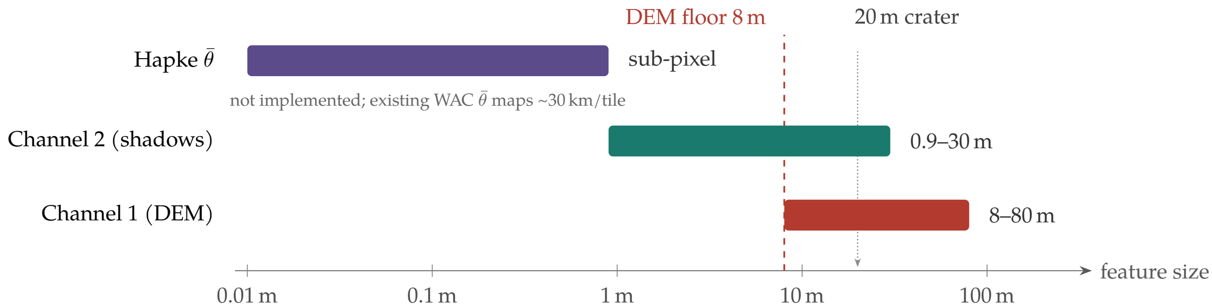

Figure 11: The three roughness measurements sit in different size bands. Channels 1 and 2 cover 8–80 m and 0.9–∼30 m. Hapke $\bar{\theta}$ covers the sub-pixel band neither touches, and the only existing lunar $\bar{\theta}$ maps are spatially coarse (∼30 km tiles). So a $\bar{\theta}$ channel is not redundant with HATI; it would add an independent, finer measurement.

Honest limit

Two things not to oversell. First, $\bar{\theta}$ is *not* redundant with what was built: it reads a sub-pixel band (fig. 11) that neither channel touches, and you cannot even fit it from the single frame used here, because the inversion needs multi-angle photometry. "Not implemented" is partly "not implementable from this data," which makes the geometric result more impressive, not less. Second, calling the result a "characteristic signal *of hazards*" is a specificity claim: shown at one hazard, not yet shown to stay quiet over safe ground. Until the control of section 9 runs, the honest phrasing is "a strong, two-channel signal at a known hazard."

8 Scrutiny: where a hostile reviewer would push

"96th percentile of a terrible neighbourhood is not an absolute hazard." Correct, and the biggest limitation. H is a within-scene ranking; the fix is a safe control site (section 9). The relative claim stands; the absolute one waits.

"You tried five settings; that is fishing." All five are reported, including the one that fails outright. The flag holds from the 93rd to the 97th percentile across every computable setting, on literature weights not tuned to Athena.

"$n = 1$, no controls." True. One positive site shows the method *can* fire on a known-bad spot; it says nothing yet about false alarms.

"Fine-scale roughness on a stereo model might be matching noise." The sharp one: a stereo model has correlated error at the few-pixel scale, exactly where the flag leans. Checked (section 9): the product confidence map (Henriksen et al. 2017) puts the touchdown at the maximum confidence value (14 of 14), level with the safe-reference patch and about 98.5 % of the valid DEM, so the flag is not sitting in a poorly-correlated patch. That confidence code is coarse and near-saturated, so it excludes a gross mismatch rather than bounding fine vertical precision; the clean, stereo-free confirmation is multi-angle shadow relief (section 9).

"Circularity." I knew the site failed. The guardrails keep the *measurement* clean, but the *site selection* is hindsight, which is why this is framed as detectability, not prediction.

9 Future work, and why each step matters

Control 1: safe-ground baseline — in-scene done, external mare next

The in-scene form of this control is already run (section 4.6): it names the channels that separate the hazard from the smoothest nearby ground. What it cannot give is the *absolute* safe floor, because the reference is the smoothest highland patch, not a mare. The external mare control is the next step, and the trap to avoid is the one section 4.6 sidesteps: it must match the 4 m/px resolution and compare raw (or commonly-normalised) values, never per-scene z, since roughness is scale-dependent (Cai and Fa 2020) and z is scene-relative. Done that way, safe-ground values far below the touchdown's would turn "rougher than 96 % of Mons Mouton" into "objectively unsafe."

Control 2: the stereo-confidence check — done

Run. The `NAC_DTM_NOBILE03_CONF` map (Henriksen et al. 2017) puts the touchdown at the maximum confidence (14 of 14), the same value as the safe-reference patch and about 98.5 % of the valid DEM, so the fine-scale roughness is not a low-confidence artifact. The remaining caveat is that the code is coarse and near-saturated: it excludes a gross stereo mismatch but does not bound fine vertical precision, which is why the stereo-free cross-check below matters.

Multi-angle shadow relief (Channel 2++)

A single image gives obstacle footprints but needs an assumed sun elevation for height. Two or more NAC frames of the site at different solar geometry let the heights be solved directly (the geometry is known per frame from SPICE) and cross-checked: a real cast shadow rotates and lengthens predictably as the sun moves, while noise and albedo do not. It is *stereo-independent*, so it is the natural confirmation of the confidence caveat in Control 2, and the multi-angle stack it needs is the same data a native-resolution $\bar{\theta}$ channel would require.

A native-resolution $\bar{\theta}$ channel

The existing $\bar{\theta}$ maps are too coarse (fig. 11) to help at a landing site. Fitting $\bar{\theta}$ from multi-angle NAC photometry would add an independent sub-pixel channel, the one band the current two miss.

Coupling to terrain-relative navigation

A hazard map only matters if a lander can act on it. The end goal is to feed H and the obstacle census into the terrain-relative navigation loop as a landing-site selection prior, so the descent avoids the worst few percent rather than reading it in hindsight.

10 Reproduce it exactly

All paths under `HATI V2.0/`. Inputs: `data/athena/NAC_DTM_NOBILE03.TIF` (4 m/px) and `..._M1101075756_90CM.IMG` (0.9 m/px). Code: `scripts/athena_counterfactual.py` (both channels), `scripts/athena_figure.py` (fig. 10), `scripts/athena_channel_influence.py` (table 1, fig. 5), and `scripts/athena_safe_control.py` (fig. 7). Run `python scripts/athena_counterfactual --stage both`, then the figure scripts. Fixed constants (no per-run tuning): touchdown 84.7906° S, 29.1957° E; elevation pixel (1181, 387); photo pixel (5248, 1721); MDS weight ladder $1.0/0.9/0.8/0.6$ by ascending size; bias -1.0; z-clip ± 4; shadow threshold $0.5\times$ local median (40 m window); obstacle floor 8 m; density radius 25 m.

11 Conclusion

Using maps that predate the landing by thirteen years, and untuned weights from the roughness literature, HATI's first channel ranks the actual IM-2 Athena touchdown in the worst few percent of an already-dangerous site, driven by fine-scale texture that a slope check misses, and its second channel confirms a dense obstacle layer below the planning model's resolution. The site was detectable. A within-scene safe-ground control already names the discriminating channels; a resolution-matched mare baseline and a stereo-confidence check would harden "worst few percent here" into "objectively unsafe." The result asks for neither more nor less than it earns: a hazard read from old data, by geometry alone, at the place a mission was lost.

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